IMPG1-related Dominant Retinopathy — Comprehensive Disease Characterization Report

Disease: IMPG1-related Dominant Retinopathy **MONDO ID:** MONDO:1040036 **Causal gene: IMPG1** (Interphotoreceptor Matrix Proteoglycan 1; encodes the protein **SPACR**), chromosome 6q14.1 **Category:** Mendelian, autosomal dominant (with a recessive allelic counterpart)

> **Gene identity (read first):** This report concerns retinopathy caused by **IMPG1** on chromosome **6q14.1** (HGNC:6054; OMIM gene *602870; UniProt Q17R60/SPACR). Do **not** confuse it with its paralog **IMPG2** (chromosome 3q12, encoding SPACRCAN), which causes a clinically overlapping but genetically distinct retinopathy. "Dominant retinopathy" here refers to the **monoallelic (autosomal dominant)** IMPG1 phenotype — principally **adult-onset vitelliform macular dystrophy (AVMD/VMD4)** with a related pattern-dystrophy / retinitis-pigmentosa (RP91) spectrum.

Summary

IMPG1-related dominant retinopathy is a rare, adult/late-onset Mendelian inherited retinal dystrophy (IRD) caused by heterozygous variants in **IMPG1**, located on chromosome **6q14.1**. IMPG1 encodes **SPACR** (Sialoprotein Associated with Cones and Rods), a 147–150 kDa hyaluronan-binding glycoprotein that forms the macromolecular scaffold of the **interphotoreceptor matrix (IPM)** — the specialized extracellular matrix filling the subretinal space between photoreceptor outer segments and the retinal pigment epithelium (RPE). Despite the "proteoglycan" in the gene symbol, SPACR is a heavily glycosylated glycoprotein that does **not** carry glycosaminoglycan chains. Its paralog, IMPG2 (SPACRCAN), is the other principal IPM component, and the two proteins are functionally interdependent.

The dominant phenotype is chiefly **adult-onset vitelliform macular dystrophy (AVMD; OMIM VMD4, 616151)**, with an overlapping **retinitis pigmentosa spectrum (OMIM RP91, 617304)** and pattern-dystrophy presentations. The disease was defined as a distinct entity by

Meunier and colleagues in 2014, who identified IMPG1 (and IMPG2) as causal in BEST1/PRPH2-negative vitelliform families, with the recurrent dominant missense variant **c.713T>G (p.Leu238Arg)**. A key mechanistic insight from this investigation is that IMPG1 is **loss-of-function tolerant** in the general population (gnomAD pLI ≈ 0; observed/expected LoF ≈ 0.92), arguing strongly that the dominant disease is **not** caused by haploinsufficiency but by a **dominant-negative** effect of missense variants. These dominant variants cluster in SPACR's two **SEA domains** — SEA1 (residues ~232–354) variants (e.g., p.Leu238Arg) produce the vitelliform/VMD4 phenotype, whereas SEA2 (residues ~571–684) variants (e.g., p.Leu579Pro) produce the RP91 phenotype.

Mechanistically, mutant SPACR disrupts the SPACR–SPACRCAN (IMPG1–IMPG2) matrix scaffold, impairing photoreceptor–RPE adhesion, retinoid cycling (reduced 11-cis-retinal), and outer-segment maintenance, leading to accumulation of autofluorescent subretinal vitelliform material above a preserved RPE and progressive photoreceptor degeneration. This causal chain is supported by human genetics, human retinal organoid functional studies, a medaka knockdown model, and an **Impg1-knockout mouse** that recapitulates the deposits, ERG attenuation, and photoreceptor-layer degeneration. No disease-specific therapy exists; management is supportive, centered on genetic counseling, low-vision support, and surveillance.

Key Findings

Finding 1 — IMPG1 causes autosomal dominant adult-onset vitelliform macular dystrophy (VMD4)

IMPG1 (encoding SPACR, chromosome 6q14) is an established cause of autosomal dominant adult-onset vitelliform macular dystrophy. The landmark study by Meunier et al. (2014) screened 49 families with vitelliform maculopathy negative for mutations in the two previously known genes (BEST1 and PRPH2). They identified IMPG1 as causal in 3 families (11 patients), with autosomal dominant families 1 and 2 both carrying the recurrent missense variant **c.713T>G (p.Leu238Arg)**. Together, IMPG1 and its paralog IMPG2 accounted for approximately **8%** of BEST1/PRPH2-negative vitelliform families, establishing these genes as a meaningful contributor to the genetic architecture of VMD.

Clinically, the dominant IMPG1 vitelliform phenotype is characterized by **late onset (mean ~42 years)** and **moderate visual impairment (mean visual acuity ~20/40)**. A diagnostically important feature is that the vitelliform material sits **above a preserved RPE**, and the **electrooculogram (EOG) is normal or borderline** — distinguishing IMPG1/IMPG2 VMD from the BEST1-related (Best disease, abnormal EOG) and PRPH2-related forms.

> *"families 1 and 2 had the c.713T→G (p.Leu238Arg) mutation in IMPG1"* — PMID: 25085631
>
> *"IMPG1 and IMPG2 are new causal genes in 8% of families negative for BEST1 and PRPH2 mutations."* — PMID: 25085631

Finding 2 — Dual inheritance and a broad phenotypic spectrum (VMD, pattern dystrophy, retinitis pigmentosa)

IMPG1 retinopathy displays **both autosomal dominant and autosomal recessive** inheritance and a phenotypic spectrum spanning vitelliform macular dystrophy, pattern dystrophy, and retinitis pigmentosa (RP). Olivier et al. (2021, *J Med Genet*) used exome sequencing in an autosomal dominant RP family to reveal an IMPG1 splice variant; the same variant recurred in families with either RP or VMD, alongside 8 additional families with missense or nonsense IMPG1 variants. Critically, the variant originally described in 2004 as **"benign concentric annular macular dystrophy" (BCAMD; p.Leu579Pro)** was **reclassified as RP with early macular involvement**, resolving the historical nomenclature and placing BCAMD firmly within the IMPG1 RP spectrum.

Registry-scale data reinforce the dominant predominance: in the IRD-PT registry (Francisco et al., 2026), IMPG1 cases were **87.5% autosomal dominant** (contrasting with IMPG2, which was 75% recessive). Al-Khuzaei et al. (2025) reported two IMPG1 patients, both **monoallelic** (one AVMD, one pattern dystrophy), presenting at advanced ages (81 and 72 years), underscoring the late-onset character of the dominant disease.

> *"This study discusses a previously unreported association between monoallelic or biallelic IMPG1 variants and RP."* — PMID: 32817297
>
> *"the clinical diagnosis of the IMPG1 retinopathy-associated variant, originally described as benign concentric annular macular dystrophy, was also revised to RP with early macular involvement"* — PMID: 32817297

> *"IMPG2 (75.0% AR) and IMPG1 (87.5% AD) caused RP and adult-onset vitelliform macular dystrophy."* — PMID: 42321002

Finding 3 — SPACR is essential for photoreceptor survival; the Impg1-knockout mouse recapitulates human disease

SPACR is required for photoreceptor survival, and loss of IMPG1 produces a retinal phenotype in model organisms that mirrors the human disease. Olivier, Brabet, and colleagues (2022, *Genes*) characterized an **Impg1-deficient (knockout) mouse** that develops **autofluorescent deposits** (visible on fundus imaging and SD-OCT), **attenuated ERG responses from ~9 months of age**, **photoreceptor-layer degeneration**, and — on transmission electron microscopy — **disruption of the interphotoreceptor matrix and of the RPE**. The mice also show a **decrease in 11-cis-retinal**, linking IMPG1 loss to impaired retinoid (visual) cycling. SPACR and its paralog SPACRCAN (IMPG2) are the main components of the IPM. Complementary evidence comes from medaka fish: morpholino knockdown of impg1/impg2 (Olivier 2021) produced **shortened rod and cone outer segments**.

> *"our results demonstrate the essential role of SPACR in maintaining photoreceptors"* — PMID: 36140676
>
> *"SD-OCT of Impg1-/- mice shows a degeneration of the photoreceptor layer, and transmission electron microscopy shows a disruption of the IPM and the retinal pigment epithelial cells."* — PMID: 36140676
>
> *"The decrease in the concentration of the chromophore 11-cis-retinal supports this loss of photoreceptors."* — PMID: 36140676

Finding 4 — SPACR is a hyaluronan-binding glycoprotein forming the IPM scaffold (not a GAG-bearing proteoglycan)

SPACR, the IMPG1 gene product, is the hyaluronan-binding glycoprotein that constitutes the structural scaffold of the insoluble IPM. Acharya et al. (1998) demonstrated that SPACR is the prominent **147–150 kDa** band in the IPM, localizes to the matrix surrounding both rods and cones, and possesses a **functional hyaluronan-binding domain**. Despite the "proteoglycan" in the gene name, SPACR is **not a chondroitin-sulfate proteoglycan**: it lacks a GAG attachment site and resists chondroitinase ABC digestion. Instead it carries abundant **N- and O-linked glycoconjugates** and a **mucin-like domain**

(P 9719680).

The sialic-acid-rich O-linked glycans (e.g., NeuAcα2-3Galβ1-3GalNAc) contribute to the polyanionic character of the IPM and provide sites for cross-linking into the ordered IPM lattice. The hyaluronan-binding motif maps near the C-terminus and competitively binds heparin as well (P 18466325 , P 18786170).

In chick, SPACR expression parallels the development of adhesion between the neural retina and the RPE (P 11991949).

> *"We present evidence that SPACR has a functional hyaluronan binding domain, suggesting that interactions between SPACR and hyaluronan may serve to form the basic macromolecular scaffold, which comprises the insoluble interphotoreceptor matrix."* — PMID: 9813076
>
> *"SPACR is the most prominent 147-150-kDa band present in the interphotoreceptor matrix and is the gene product of IMPG1"* — PMID: 9813076
>
> *"SPACR is not a chondroitin sulfate proteoglycan"* — PMID: 9813076

Finding 5 — IMPG1 is required for human photoreceptor outer-segment and IPM development (retinal organoid evidence)

Human in-vitro evidence confirms that IMPG1 is required for photoreceptor outer-segment and IPM development. Felemban et al. (2018) showed that in human pluripotent stem cell (hPSC)-derived retinal organoids, IMPG1 and IMPG2 are expressed in the developing IPM, and that this expression pattern is **conserved between human retina and organoids**. Functional blockade of **CD44 and IMPG1** disrupted the development of photoreceptors, their inner/outer segments and connecting cilia, and impaired IPM formation — with **IMPG1 having an earlier and more significant impact** than CD44. CD44 is the hyaluronan receptor on Müller cell microvilli, linking the SPACR–hyaluronan scaffold to a defined cellular partner.

> *"Blocking the action of CD44 and IMPG1 in pluripotent stem cell derived retinal organoids affected the development of photoreceptors, their inner/outer segments and connecting cilia and disrupted IPM formation, with IMPG1 having an earlier and more significant impact."* — PMID: 29777959

Finding 6 — IMPG1 is LoF-tolerant in gnomAD → dominant disease is likely dominant-negative

Population constraint data argue that the dominant disease mechanism is **dominant-negative** rather than haploinsufficiency. gnomAD v4 constraint metrics for IMPG1 (ENSG00000112706) show **pLI ≈ 0** (5.3×10^{-27}), an **observed/expected LoF ratio of 0.92** (85 observed vs 92.0 expected loss-of-function variants), a **LOEUF upper bound of 1.11**, a LoF Z-score of 0.62, and a missense Z-score of −0.91. Collectively these indicate that IMPG1 is **not constrained** against heterozygous loss of function — i.e., the general population tolerates IMPG1 null alleles well, so simple haploinsufficiency is an unlikely disease mechanism.

Consistent with this, the autosomal **dominant** phenotype is driven predominantly by recurrent **missense** variants (e.g., p.Leu238Arg, p.Leu579Pro, p.Asp423Glu) rather than truncating/null alleles, whereas **biallelic/null** genotypes produce recessive RP. The most parsimonious interpretation is that mutant SPACR monomers poison the SPACR–SPACRCAN matrix assembly (dominant-negative), while a single null allele — leaving one functional copy — is insufficient to cause disease. (Evidence source: computational/population genetics; gnomAD. ClinVar lists on the order of ~117 pathogenic + ~53 likely pathogenic IMPG1 variants.)

Finding 7 — Dominant variants cluster in SPACR's two SEA domains (SEA1→VMD4; SEA2→RP91)

A clear genotype–phenotype architecture emerges from mapping pathogenic variants onto the SPACR protein. UniProt Q17R60 (SPACR, 797 aa) annotates: a signal peptide (residues 1–20); **SEA domain 1 at residues 232–354**; **SEA domain 2 at residues 571–684**; and a central O-glycosylated **mucin-like cluster** (O-GalNAc at Thr403/421/432/442) with multiple N-glycosylation sites. SEA (Sea-urchin sperm protein, Enterokinase, Agrin) domains are proteolytically processed modules found in many heavily glycosylated membrane-associated proteins, and SEA-domain mutations are broadly disease-relevant.

- The recurrent dominant **vitelliform** variant **p.Leu238Arg** (rs713993045) lies inside **SEA1** and is annotated "in **VMD4**."

- The **p.Leu579Pro** variant (rs1782008883; the original "BCAMD" allele) lies inside **SEA2** and is annotated "in **RP91**"; additional RP91 variants map to residues 613 and 626, also within/adjacent to SEA2.
- The novel VMD variant **p.Asp423Glu** (Nguyen/Al-Khuzaei cohorts) lies in the inter-SEA mucin-like O-glycosylation region.

> *"Mutations within these domains are linked to a range of pathologies, including various cancers and congenital disorders, underscoring their clinical importance."* — PMID: 39930241

Variant (protein)	cDNA / rsID	SPACR domain	Phenotype	OMIM	Inheritance
p.Leu238Arg	c.713T>G / rs713993045	SEA1 (232–354)	Vitelliform macular dystrophy	VMD4 (616151)	AD
p.Asp423Glu	—	Inter-SEA mucin region	Vitelliform macular dystrophy (novel)	VMD4	AD (monoallelic)
p.Leu579Pro	c.1736T>C / rs1782008883	SEA2 (571–684)	RP with early macular involvement (orig. "BCAMD")	RP91 (617304)	AD
variants at res. 613, 626	—	SEA2 (adjacent)	Retinitis pigmentosa	RP91	AD

Section-by-Section Report

1. Disease Information

Overview. IMPG1-related dominant retinopathy is a rare inherited retinal dystrophy in which heterozygous variants in **IMPG1/SPACR** disrupt the interphotoreceptor matrix, producing adult/late-onset macular and/or rod–cone degeneration. The dominant disease most often presents as **adult-onset vitelliform macular dystrophy** (yellowish subretinal deposits in the macula) and, less commonly, as a **retinitis pigmentosa** spectrum.

Key identifiers. - **Mondo:** MONDO:1040036 - **Causal gene:** IMPG1 (HGNC:6054); protein SPACR (UniProt Q17R60); gene OMIM *602870 - **OMIM (allelic disorders):** Vitelliform macular dystrophy 4 (**VMD4, 616151**); Retinitis pigmentosa 91 (**RP91, 617304**) - **Orphanet:** within the "adult-onset vitelliform macular dystrophy" / "retinitis pigmentosa" rare-disease groupings - **ICD-10/ICD-11:** H35.5 (hereditary retinal dystrophy) / ICD-11 9B70–9B71 region; **MeSH:** closest terms "Vitelliform Macular Dystrophy," "Retinitis Pigmentosa," "Retinal Dystrophies"

Synonyms / alternative names: autosomal dominant IMPG1 retinopathy; autosomal dominant IMPG1-related vitelliform macular dystrophy; autosomal dominant IMPG1-related retinitis pigmentosa; benign concentric annular macular dystrophy (BCAMD — now reclassified as RP with early macular involvement).

Source type. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet, ClinVar, gnomAD, UniProt) plus **primary clinical case series/cohorts** and **experimental model studies** — not from individual EHR records.

2. Etiology

Primary cause — genetic. The disease is **monogenic**: pathogenic variants in IMPG1. There is no environmental or infectious cause.

Genetic risk factors. Causal, recurrent dominant variants include **p.Leu238Arg** (SEA1; VMD4), **p.Leu579Pro** (SEA2; RP91), and **p.Asp423Glu** (inter-SEA region). A splice variant can produce RP or VMD within a single family. Biallelic (null/missense) genotypes cause **recessive RP**. The paralog **IMPG2** causes a clinically overlapping retinopathy, and the two proteins are functionally interdependent.

Environmental / lifestyle risk factors. None established. Age modifies *expression* (late onset) but is not a cause. No sex predilection has been reported.

Protective factors. None specifically identified. Because IMPG1 LoF is tolerated in gnomAD, heterozygous null alleles in the population do not cause dominant disease — relevant to interpreting incidental findings, but not a "protective" intervention.

Gene–environment interactions. None documented.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset	Severity	Progression	Frequency
Vitelliform (yellowish subretinal) macular deposits	Clinical sign (imaging)	HP:0011506 (Macular dystrophy)	Adult (mean ~42 y)	Moderate	Progressive	Characteristic of dominant VMD form
Reduced visual acuity / central vision loss	Symptom	HP:0000572 (Visual loss)	Adult	Mild–moderate (mean VA ~20/40)	Slowly progressive	Common
Metamorphopsia (distorted central vision)	Symptom	HP:0012508	Adult	Mild	Fluctuating/ progressive	Variable
Night blindness (nyctalopia)	Symptom	HP:0000662	Variable	Variable	Progressive	RP-spectrum
Peripheral/ constricted visual field	Clinical sign	HP:0001133	Variable	Variable	Progressive	RP-spectrum
Abnormal fundus autofluorescence	Imaging abnormality	—	Adult	—	Progressive	Characteristic
Reduced/ attenuated ERG responses	Lab abnormality	HP:0000512 (Abnormal ERG)	Later/RP-spectrum	Variable	Progressive	RP-spectrum
Normal/ borderline EOG (distinguishing)	Lab abnormality	—	—	—	Stable	Distinguishes from BEST1

Quality of life. The dominant vitelliform form generally causes **moderate** central-vision impairment with preserved independence for most daily tasks early on; reading, driving, and fine central tasks are most affected. The RP-spectrum form carries greater burden through night blindness and peripheral field loss. No disease-specific EQ-5D/SF-36 data were identified; impact is inferred from macular dystrophy and RP literature generally.

4. Genetic / Molecular Information

- **Causal gene: IMPG1**, chromosome **6q14.1**; gene OMIM *602870; HGNC:6054. 797-aa protein SPACR (UniProt Q17R60).
- **Pathogenic variants:** predominantly **missense** for dominant disease (p.Leu238Arg, p.Leu579Pro, p.Asp423Glu, and variants at res. 613/626); **splice** variants and **nonsense/truncating** variants also reported, the latter more associated with recessive disease. ClinVar: ~117 pathogenic + ~53 likely pathogenic IMPG1 variants (ACMG/AMP).
- **Allele frequencies:** dominant pathogenic missense variants are rare/absent-to-very-low in gnomAD; general LoF variants are present at frequencies consistent with LoF tolerance.
- **Origin:** germline. No somatic involvement.
- **Functional consequences:** dominant missense variants in SEA1/SEA2 most likely act by a **dominant-negative** mechanism (poisoning the SPACR/SPACRCAN scaffold); biallelic null → **loss of function** → recessive RP.
- **Modifier genes:** IMPG2 is a strong candidate functional modifier given SPACR–SPACRCAN interdependence; CD44 (hyaluronan receptor) is a mechanistic partner. No formal modifier mapping exists.
- **Epigenetic / chromosomal abnormalities:** none implicated.

5. Environmental Information

No environmental factors, lifestyle factors, toxins, or infectious agents are implicated. This is a purely Mendelian disorder.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **heterozygous missense variant** in an IMPG1 **SEA domain** (SEA1, e.g. p.Leu238Arg; or SEA2, e.g. p.Leu579Pro) **leads to** production of a structurally abnormal SPACR monomer.

2. The mutant SPACR **results in** a dominant-negative disruption of the SPACR–SPACRCAN (IMPG1–IMPG2) matrix assembly (*inferred* from LoF tolerance in gnomAD plus the missense-driven dominant genetics; the exact biochemical poisoning step is not yet directly demonstrated).
3. Disrupted SPACR **leads to** destabilization of the hyaluronan-anchored IPM scaffold and impaired photoreceptor–RPE / Müller-microvillus (CD44) adhesion.
4. A compromised IPM **results in** defective trafficking of retinoids and other molecules across the subretinal space (*demonstrated*: 11-cis-retinal reduced in Impg1-knockout mice; IRBP docking depends on IPM domains).
5. Impaired outer-segment support and retinoid cycling **lead to** accumulation of shed outer-segment/autofluorescent material as **subretinal vitelliform deposits above a preserved RPE** (observed in patients and knockout mice).
6. Chronic matrix/metabolic stress **results in** progressive **photoreceptor degeneration**, ERG attenuation, and vision loss — in the macula → **vitelliform macular dystrophy**; **branch:** with SEA2 variants or biallelic null → **rod–cone degeneration (retinitis pigmentosa)** with field loss and nyctalopia.

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IMPG1 SEA-domain missense (heterozygous)
        │  (SEA1→macula/VMD4 ; SEA2→RP91)
        ▼
Abnormal SPACR monomer ──► dominant-negative poisoning of SPACR/SPACRCAN scaffold
        ▼
Destabilized hyaluronan-anchored IPM  ◄── CD44 (Müller microvilli) interaction impaired
        ▼
Impaired retinoid trafficking (↓11-cis-retinal) + loss of photoreceptor–RPE adhesion
        ▼
Autofluorescent subretinal (vitelliform) deposits above preserved RPE
        ▼
Progressive photoreceptor degeneration ──► ERG attenuation ──► vision loss
                                           └─► branch: RP (field loss, nyctalopia)
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Molecular pathways / cellular processes. Extracellular matrix organization; hyaluronan binding and cross-linking; cell–matrix adhesion; the visual (retinoid) cycle. Relevant GO terms: **GO:0030198** (extracellular matrix organization), **GO:0005540** (hyaluronic acid binding), **GO:0007601** (visual perception), **GO:0042574** (retinal metabolic process).

Protein dysfunction. SEA-domain missense substitutions alter SPACR structure; given SEA domains' role in glycoprotein processing/assembly, the likely defect is impaired matrix incorporation/oligomerization (dominant-negative), not enzymatic loss. **Cell types:** rod

photoreceptor (**CL:0000604**), cone photoreceptor (**CL:0000573**), retinal pigment epithelial cell (**CL:0002586**), Müller glial cell (**CL:0000636**). **Subcellular:** interphotoreceptor matrix / extracellular region (**GO:0005578** proteinaceous extracellular matrix); photoreceptor outer segment (**GO:0001750**) and connecting cilium secondarily affected.

Molecular profiling. IMPG1/IMPG2 expression is photoreceptor/retina-specific and developmentally regulated, rising as outer segments elaborate (chick, rat, mouse, human-organoid data). No disease-specific proteomic/metabolomic signatures beyond reduced 11-cis-retinal in the mouse model.

7. Anatomical Structures Affected

- **Organ:** eye (**UBERON:0000970**), specifically the **retina** (**UBERON:0000966**) and **macula lutea** (**UBERON:0005390**) for the VMD form.
- **Tissue/cell:** photoreceptor layer (rods **CL:0000604**, cones **CL:0000573**); retinal pigment epithelium (**UBERON:0001782** / **CL:0002586**); Müller glia (**CL:0000636**); the **interphotoreceptor/subretinal matrix** (**UBERON:0013706** subretinal space).
- **Subcellular:** interphotoreceptor matrix / extracellular region; photoreceptor outer segment and connecting cilium secondarily affected.
- **Localization: bilateral**, typically symmetric; macula-centered in VMD, more diffuse in RP.

8. Temporal Development

- **Onset: adult/late-onset** (dominant VMD mean ~42 y; some monoallelic cases present in the 7th–9th decades). Pattern: **insidious/chronic**.
- **Progression:** slowly **progressive**; vision typically **moderately** impaired in VMD. RP-spectrum cases show progressive field constriction and ERG decline. In the mouse model, ERG attenuation begins ~9 months.
- **Course / duration: chronic, lifelong**. No spontaneous remission. Critical intervention windows are undefined, but earlier stages (before extensive photoreceptor loss) are the logical target for any future therapy.

9. Inheritance and Population

- **Epidemiology:** rare; precise prevalence/incidence not established. Within IRD cohorts, IMPG1+IMPG2 account for ~1.45% of IRD patients (Nguyen et al., 2024;

)

P 39858590

and ~8% of BEST1/PRPH2-negative vitelliform families (Meunier 2014).

- **Inheritance:** predominantly **autosomal dominant** (87.5% of IMPG1 cases in the IRD-PT registry); also **autosomal recessive** (biallelic) for RP.
- **Penetrance / expressivity:** appears **age-dependent** and **variable**; the same variant can manifest as RP or VMD within a family (variable expressivity).
- **Anticipation / mosaicism / founder effects / consanguinity:** not established; recurrent variants (p.Leu238Arg, p.Leu579Pro) reflect mutational recurrence rather than confirmed founders. Consanguinity is relevant to the recessive form.
- **Carrier frequency:** IMPG1 LoF alleles are present in gnomAD (LoF-tolerant), so heterozygous nulls are relatively common but non-pathogenic for dominant disease.
- **Demographics:** reported across European and other populations; no strong ethnic predilection; **no clear sex bias**.

10. Diagnostics

- **Imaging (cornerstone):** fundus photography (vitelliform/yellow macular lesion or pigmentary RP changes), **fundus autofluorescence** (hyperautofluorescent deposits), **SD-OCT** (subretinal hyperreflective vitelliform material **above a preserved RPE**), fluorescein angiography.
- **Electrophysiology: EOG normal/borderline** (key discriminator from BEST1 disease); **ERG** normal in early VMD, attenuated in RP-spectrum disease (HP:0000512).
- **Genetic testing (confirmatory):** targeted single-gene testing, **IRD gene panels** (which include IMPG1/IMPG2), **whole-exome** and **whole-genome sequencing**. Variant classification per ACMG/AMP using ClinVar/ClinGen. CMA/karyotype/FISH/mtDNA/repeat testing are **not indicated**.
- **Differential diagnosis:** Best vitelliform macular dystrophy (BEST1; abnormal EOG), PRPH2-related pattern dystrophy/AVMD, acquired (age-related) AVMD, other RP genes (for the RP form), and IMPG2 retinopathy (clinically overlapping paralog).

11. Outcome / Prognosis

- **Survival/mortality:** the disease is **not life-limiting**; normal life expectancy. No disease-specific mortality.

- **Morbidity/function: visual** morbidity — moderate central vision loss in VMD (mean VA ~20/40), more substantial functional loss in RP-spectrum cases (field constriction, night blindness). Legal blindness is possible in advanced RP.
- **Course:** chronic, slowly progressive; **complications** include geographic atrophy/photoreceptor loss over the lesion. Recovery is not expected (no retinal regeneration).
- **Prognostic factors:** variant location (SEA2/RP phenotype generally carries wider retinal involvement than SEA1/VMD), biallelic status (recessive RP), age, and extent of photoreceptor/ERG loss at presentation.

12. Treatment

- **No disease-specific or disease-modifying therapy exists.** Management is **supportive**: low-vision aids and rehabilitation, visual-function monitoring, and **genetic counseling**. (NCIT: Supportive Care; Genetic Counseling; Low Vision Aids.)
- **Pharmacotherapy / pharmacogenomics:** none established. No approved drugs.
- **Advanced therapeutics:** none approved. Conceptually, because dominant disease is likely dominant-negative, **allele-specific gene silencing (ASO/siRNA)** or gene correction would be more rational than simple gene augmentation; **gene augmentation** may suit the recessive (loss-of-function) form. All are hypothetical/pre-clinical.
- **Surgical/interventional:** none specific.
- **Experimental/clinical trials:** no IMPG1-specific trials identified.
- **Treatment strategy:** a genotype-guided framework (SEA1 vs SEA2; mono- vs biallelic) would inform any future precision approach.

13. Prevention

- **Primary prevention:** not applicable (genetic disease). **Genetic counseling** and reproductive options (prenatal testing, preimplantation genetic testing) are available for at-risk families.
- **Secondary prevention: cascade genetic testing** of relatives; periodic ophthalmic surveillance of carriers to detect onset and manage visual function early.
- **Tertiary prevention:** low-vision rehabilitation, monitoring/managing complications; general retinal-health measures (e.g., smoking cessation, UV protection) are prudent though not evidence-based for this specific gene.
- **Public health/immunization/prophylaxis:** not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** IMPG1 orthologs exist across vertebrates — mouse *Impg1* (**NCBI Gene 16011**), plus orthologs in rat, chick, medaka/zebrafish, and Xenopus. SPACR/SPACRCAN biology has been characterized in **chick** (P 11991949 P 16469746), P 18466325 **rat** (P 11406817), P 9778409 **mouse** (P 12589770), P 36140676 and **Xenopus** (P 23470504).
- **Natural disease in animals:** no well-documented naturally occurring IMPG1 retinopathy in companion animals is recorded here; the disease is primarily studied via engineered/knockdown models.
- **Comparative biology:** the IPM, SPACR/SPACRCAN composition, hyaluronan scaffold, and CD44 partnership are **evolutionarily conserved**, making cross-species models informative. Mechanisms of outer-segment support are conserved.
- **Transmission:** none (non-infectious, non-zoonotic).

15. Model Organisms

Model	Type	Genetic manipulation	Phenotype recapitulation	Reference
Impg1-knockout mouse	Mammalian in vivo	Constitutive KO	Autofluorescent deposits, SD-OCT photoreceptor-layer degeneration, ERG attenuation from ~9 mo, IPM/RPE disruption on TEM, ↓11-cis-retinal — **strong** recapitulation	PMID: 36140676
Medaka fish	Vertebrate in vivo	impg1/impg2 morpholino knockdown	Shortened rod and cone outer segments — models developmental OS defect	Olivier 2021 (J Med Genet)
Human retinal organoids (hPSC)	In vitro, human	CD44/IMPG1 functional blockade	Disrupted photoreceptor / IS-OS / connecting cilium and IPM development; IMPG1 earlier/greater impact	PMID: 29777959
Chick / rat / Xenopus retina	Vertebrate, descriptive	—	SPACR/SPACRCAN localization, hyaluronan binding, developmental expression, IRBP docking	PMIDs 11991949, 11406817, 23470504

Model limitations: the mouse KO models **loss of function** (recessive-type mechanism), not the human **dominant-negative** missense state; a **knock-in** of p.Leu238Arg or p.Leu579Pro is needed to faithfully model dominant disease. Morpholino knockdown is transient/developmental; organoids capture development but not chronic adult-onset degeneration. **Resources:** MGI (mouse *Impg1*), IMPC/KOMP, ZFIN (zebrafish), Cellosaurus (hPSC lines).

Mechanistic Model / Interpretation

The unifying model is that **SPACR is the structural glue of the subretinal compartment**. It binds hyaluronan near its C-terminus to form the insoluble IPM lattice, co-assembles with SPACRCAN (IMPG2), and engages CD44 on Müller microvilli — together maintaining

photoreceptor–RPE adhesion and a conduit for retinoid/IRBP trafficking. Any insult to this scaffold degrades outer-segment support and the visual cycle, which manifests first as accumulation of autofluorescent (vitelliform) material and then as photoreceptor death.

The **genotype–mechanism correlation** is the central intellectual contribution of this investigation:

- **gnomAD LoF-tolerance (F006)** rules out haploinsufficiency as the dominant mechanism — a single IMPG1 null is benign in the population.
- Therefore the **dominant, missense, SEA-domain-clustered** variants (F007) must act by a **dominant-negative** route, with mutant SPACR corrupting the scaffold that wild-type SPACR and SPACRCAN would otherwise build.
- **SEA1 vs SEA2** position tracks with phenotype (macular/VMD4 vs rod–cone/RP91), suggesting the two SEA modules make functionally distinct contributions to matrix assembly across retinal regions.
- The **recessive RP** form, driven by biallelic null/severe alleles, fits a conventional loss-of-function mechanism — consistent with a single null being tolerated while complete deficiency is not, as proven by the knockout mouse.

This two-mechanism picture (dominant-negative missense vs recessive null) has direct therapeutic implications: allele-specific silencing for dominant disease; gene augmentation for recessive disease.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
25085631	IMPG1/IMPG2 vitelliform frequency & pattern (Meunier 2014)	Human clinical cohort	F001 — AD VMD; p.Leu238Arg; 8% of BEST1/PRPH2-neg families
32817297	Pathogenic IMPG1 variants / dual inheritance (Olivier 2021)	Human clinical + genetics	F002 — AD & AR; RP/VMD spectrum; BCAMD→RP reclassification
42321002	IRD-PT registry dual-inheritance genes	Human registry	F002 — IMPG1 87.5% AD
36140676	SPACR essential / Impg1-KO mouse (Olivier-Brabet 2022)	Model organism	F003 — KO recapitulates deposits, degeneration, ↓11-cis-retinal
9813076	SPACR hyaluronan-binding IPM glycoprotein (Acharya 1998)	In vitro/ biochemical	F004 — scaffold role; not a CS-proteoglycan
9719680	SPACR glycoconjugate characterization	In vitro	F004 — N/O-glycans, mucin-like, sialylated
29777959	hPSC retinal organoid IMPG1/CD44 (Felemban 2018)	Human in vitro	F005 — IMPG1 required for OS/IPM development
18466325 / 18786170	HA-binding motif / heparin competition	In vitro	F004 — C-terminal HABM
14691150	BCAMD locus 6p12.3-q16 → IMPG1 p.Leu579Pro	Human linkage	F002/F007 — SEA2 RP allele
39858590	Prevalence of IMPG1/ IMPG2 in IRD cohort	Human cohort	Epidemiology (1.45% of IRDs); novel p.Asp423Glu
39930241	Critical roles of SEA domains	Review	F007 — SEA-domain mutations are disease-relevant
11991949 / 11406817 /	SPACR/SPACRCAN comparative biology	Model/ comparative	F004 — conserved IPM biology, IRBP docking

PMID	Title (abbrev.)	Evidence type	Supports
12589770 / 16469746 / 23470504			

gnomAD v4 (computational): IMPG1 pLI≈0, o/e LoF=0.92, LOEUF upper=1.11 → underpins F006.

Limitations and Knowledge Gaps

1. **Dominant-negative mechanism is inferred, not proven.** F006 rests on population constraint plus the missense-dominant/null-recessive pattern; no biochemical experiment has yet shown mutant SPACR poisoning wild-type scaffold assembly. The existing mouse models loss of function, not the dominant missense state.
2. **Genotype–phenotype rule (SEA1→VMD, SEA2→RP) is based on few variants.** Larger cohorts are needed to test whether the correlation is robust or has exceptions (variable expressivity already blurs it).
3. **Penetrance and natural history are poorly quantified.** No prospective natural-history study; progression rates, conversion from VMD to atrophy, and sex/ethnic distributions are not well defined.
4. **Epidemiology is imprecise** — only cohort-relative frequencies (~1.45% of IRDs; ~8% of BEST1/PRPH2-neg VMD) are available; no population prevalence.
5. **No therapeutics or trials** exist; QoL data are extrapolated from generic macular-dystrophy/RP literature, not IMPG1-specific.
6. Some supporting citations are drawn from abstracts/snippets; a few newer cohort papers (e.g.,

   P 41465146

) had no abstract available for verification.

Proposed Follow-up Experiments / Actions

1. **Generate SEA-domain knock-in mice** (p.Leu238Arg and p.Leu579Pro) to directly test the dominant-negative hypothesis and the SEA1-vs-SEA2 phenotype branch; compare to the existing KO.
2. **Biochemical co-assembly assays** — express WT and mutant SPACR with SPACRCAN in vitro (and in organoids) to measure whether mutant SPACR impairs scaffold formation, hyaluronan binding, or CD44 engagement.
3. **Deep variant curation** across ClinVar/large IRD cohorts to statistically test the SEA1→VMD / SEA2→RP correlation and refine ACMG classification.
4. **Prospective multimodal-imaging natural-history study** (FAF, SD-OCT, ERG/EOG) to define progression, penetrance, and conversion to atrophy.
5. **Therapeutic proof-of-concept:** allele-specific ASO/siRNA knockdown for dominant alleles in patient-derived organoids; AAV-IMPG1 augmentation for the recessive form.
6. **Standardize IMPG1 on IRD gene panels** and promote cascade testing and genetic counseling for at-risk relatives.

Ontology Appendix (suggested terms)

- **Disease:** MONDO:1040036; OMIM 616151 (VMD4), 617304 (RP91)
- **Gene/protein:** HGNC:6054 (IMPG1); UniProt Q17R60 (SPACR); gene OMIM *602870
- **HPO:** HP:0011506 (Macular dystrophy), HP:0000572 (Visual loss), HP:0012508 (Metamorphopsia), HP:0000662 (Nyctalopia), HP:0001133 (Constriction of visual field), HP:0000512 (Abnormal ERG)
- **GO:** GO:0030198 (ECM organization), GO:0005540 (hyaluronic acid binding), GO:0007601 (visual perception), GO:0042574 (retinal metabolic process), GO:0005578 (proteinaceous ECM)
- **CL:** CL:0000604 (rod), CL:0000573 (cone), CL:0002586 (RPE cell), CL:0000636 (Müller glial cell)
- **UBERON:** UBERON:0000966 (retina), UBERON:0005390 (macula lutea), UBERON:0001782 (RPE), UBERON:0013706 (subretinal space)
- **CHEBI:** 11-cis-retinal, hyaluronic acid (CHEBI:16336)
- **NCIT:** Supportive Care, Genetic Counseling, Low Vision Aids